Neighborly polytopes

A polytope is k-neighborly if every nonempty subset of k or fewer vertices is the vertex set of a face.

1-neighborly: Every vertex is a face

Every polytope is 1-neighborly.

2-neighborly: Every two vertices form an edge of the polytope.

E.g. simplices

No other 3-dim examples, since K_5 is not planar.

Let $t_1 < t_2 < \cdots < t_n$ be real numbers. Let

$$C(d,n) = \operatorname{conv}\big((t_1, t_1^2, \ldots, t_1^d), \\ (t_2, t_2^2, \ldots, t_2^d), \\ \vdots \\ (t_n, t_n^2, \ldots, t_n^d) \big) \subset \mathbb{R}^d$$

cyclic polytope

<u>Prop</u>: $C(d,n)$ is $\lfloor d/2 \rfloor$-neighborly.

Proof: It suffices to show that any $k \le \lfloor d/2 \rfloor$ of the above points are the vertices of a face of $C(d,n)$.

Let S be a subset of the points with $|S| = k \leq \lfloor d/2 \rfloor$.

$$S = \{t_{i_1} < \cdots < t_{i_k}\}$$

Want a linear functional $(a_1, \ldots, a_d) \in (\mathbb{R}^d)^*$ such that

$$a_1 t_i + a_2 t_i^2 + \cdots + a_d t_i^d = b \quad (t_i, \; t_i^d) \in S$$

$$> b \quad \text{otherwise}$$

Let $? = -b + a_1 t_i + \cdots + a_d t_i^d$

$$? = (t - t_{i_1})^2 (t - t_{i_2})^2 \cdots (t - t_{i_k})^2$$

has degree $2k \leq d$

we choose $a_1, \ldots, a_d, b$, so that

$$-b + a_1 t + \cdots + a_d t_i^d =$$

Thus $\operatorname{conv} S$ is a face. Applied to $k=1$, each $(t_i, \ldots, t_i^d)$ is a vertex, so the points of S are the vertices of $\operatorname{conv} S$.

<u>Prop</u>: A d-polytope which is not a simplex can be at most $\lfloor d/2 \rfloor$-neighborly.

Proof: Suppose P is k-neighborly d-polytope with at least $d+2$ vertices. Pick some subset V of vertices with $d+2$ elements.

$$\sum_{x \in V} \lambda_x x = 0 \qquad \sum_{x \in V} \lambda_x = 0$$

λ_x are real numbers, not all 0.

Partition V into V^+, V^- depending on the sign of λ_x. (If $\lambda_x = 0$, assign arbitrarily.) Both V^+, V^- are nonempty. $\mu_x = -\lambda_x$

$$\sum_{x \in V^+} \lambda_x x - \sum_{x \in V^-} \mu_x x = 0$$

Scale so that $\sum_{x \in V^+} \lambda_x = \sum_{x \in V^-} \mu_x = 1$.

Let $y = \sum_{x \in V^+} \lambda_x x = \sum_{x \in V^-} \mu_x x$, then y is in the convex hull of V^+ and V^-. Since $|V| = d+2$, one of these has $\leq \lfloor d/2 \rfloor + 1$ elements.

If $k \geq \lfloor d/2 \rfloor + 1$, then this subset forms a face. However, y is in this face and can be written as the convex hull of points not in the face, contradicting def of face.

A d-polytope is **neighborly** if it is $\lfloor d/2 \rfloor$-neighborly.

There are in fact many, many neighborly polytopes which are combinatorially distinct.

A **circuit** is a finite set $X \subset \mathbb{R}^d$ which is affinely dependent, and any proper subset is affinely independent.

Let X be a circuit. We can write

$$\sum_{x \in X} \lambda_x x = 0 \qquad \sum \lambda_x = 0$$

for some real $\lambda_x \neq 0$, and this relation is unique up to scaling.

Partition $X = (X^+, X^-)$ dependending on the sign of λ_x.

Prop: Let S be a finite subset of $\mathbb{R}^d$ and $T \subset S$. Then $T = S \cap F$ for some face of conv S iff there is no circuit (X^+, X^-) of S with $X^+ \subset T$, $X^- \not\subset T$.

In other words, knowing all the circuits with their associated partitions is enough to know the
(signed circuits)
combinatorial structure of conv S.

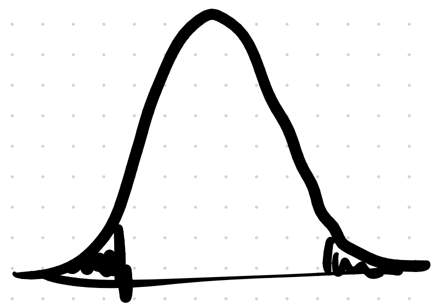
Cor: A set $S \subset \mathbb{R}^d$ is the set of vertices of a k-neighborly polytope iff it has no circuits (X^+, X^-) with $|X^+| \le k$.

$x_1, \ , \ldots, \ x_{d+2}$

$+ \qquad - \qquad +$

$$P(|X|^+ \le k \text{ or } |X|^- \le k)$$

$$\lesssim 2^{-d}$$



Prop: Let $v_1, \dots, v_n \in \mathbb{R}^d$ such that every maximal minor of

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ v_1 & v_2 & v_3 & & v_n \end{pmatrix}$$

is postive. Then $\mathrm{conv}(v_1, \dots, v_n)$ is combinatorially equivalent to $C(d,n)$.

Conversely, every polytope combinatorially equivalent to $C(d,n)$ is of this form.

Prop: The combinatorial type of $\mathrm{conv}(v_1, \dots, v_n)$ is determined by the signs of the maximal minors of

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ v_1 & v_2 & \cdots & v_n \end{pmatrix}$$